In-Service Measurements of Polarization-Mode Dispersion and Correlation to Bit-Error Rate

Misha Boroditsky, Misha Brodsky, Nicholas J. Frigo, *Member, IEEE*, Peter Magill, and L. Raddatz

***Abstract*—We introduce a nondisruptive *in situ* polarization-mode dispersion measurement technique using the sidebands of working channels and show that it compares favorably with the standard Mueller matrix method. Using this technique, we observed time correlations between the instantaneous differential group delay and bit-error rate on a 1000-km field-installed 43-Gb/s transmission system.**

***Index Terms*—Optical fiber communications, polarization-mode dispersion (PMD), wavelength-division multiplexing.**

I. Introduction

POLARIZATION-MODE dispersion (PMD) is recognized as a possible limiting impairment for high-speed fiber-optic communication systems. With the advent of 40-Gb/s or higher speed systems, the need arises for accurate PMD measurements. Since PMD changes with time, real-time PMD measurement may be necessary for compensation or mitigation purposes. In this letter, we propose such a method, demonstrate its accuracy, and report its use to measure real-time correlation between PMD and bit-error rate (BER) for a 1000-km 43-Gb/s link.

Most of the conventional PMD measurements, such as the Jones matrix eigenvalue (JME) analysis, Mueller matrix method (MMM), and Stokes parameters method require use of an external probe laser scanning across the frequency range of interest [1]. Performing such a measurement on a working system would be very intrusive to system operation and may require shutting down the channel being measured. It is sometimes feasible to scan the probe only in the spectral vicinity of the operating channel, but this approach has several shortcomings. First, the presence of the probe may distort the transmission signal and cause errors. Second, in a long amplified system, the output polarization of the probe may be influenced by the channel light though nonlinear polarization rotation [2]. Finally, scanning the probe across multiple channels will require turning it on and off, which in turn may cause transient power fluctuations in amplified systems. Therefore, it would be advantageous to use the relatively broad spectrum of the modulated optical signal as a probe without interrupting the service. One such method based on broad-band spectrum was proposed by Möller and Buhl [3]. This method requires a polarizer at the transmitter, which is not practical in a real system. In this work, we extend the MMM to use the sidebands of data-carrying channels to measure PMD while avoiding this limitation. We observe a good agreement in simultaneous measurement of the PMD in a 43-Gb/s carrier-suppressed return-to-zero (CSRZ) system using two methods. Once the *in situ* method's validity was established, we used it to measure the differential group delay (DGD) simultaneously with the BER. Even though the BER in the presence of PMD depends on the launch state of polarization (SOP), we observed good correlation between BER and PMD in an eight-channel 43-Gb/s Raman-amplified system. Generally speaking, these measurements do not need to be based on the lobes of the optical spectrum. Therefore, this method can be extended to other modulation formats, and to higher orders of PMD.

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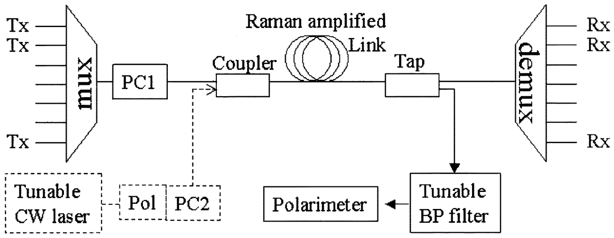
II. Experimental Setup and Results

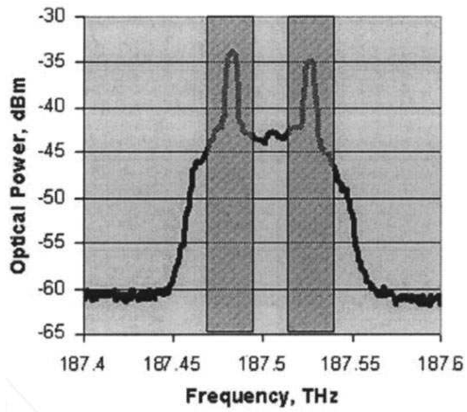
With MMM [4], DGD and principal state polarization (PSP) are calculated based on the differential rotation of a received polarization basis set. This method requires the measurement of output Stokes parameters for two launch SOPs at two adjacent frequencies. Without noise, any two SOPs (not parallel in Stokes space) define a basis set, and two SOP pairs are sufficient to determine the magnitude and relative orientation of the DGD vector and its evolution in time. In the presence of noise, one needs to launch sufficiently different SOPs, and typically a polarizer is used for precise control of the launch polarization states.

As one can see from the spectrum of a 43-Gb/s CSRZ signal, it has two strong lobes separated by $\Delta f = 43$ GHz. Our intention is to use the optical power in these lobes as the probe signal. This means that we have a frequency step of $\Delta f = 43$ GHz, which determines a maximum measurable DGD of $\tau_{\max} = 1/(2\Delta f) = 12$ ps, which is exactly half of the bit period. Certainly, this relation between the maximum DGD and bit length holds for other bit rates. In this work, we used a tunable 0.25-nm (31-GHz) optical bandpass filter to separate lobes, as indicated with shading in Fig. 1(b). Our experimental setup for the measurement using the sidebands is shown in Fig. 1(a). Eight multiplexed 43-Gb/s signals pass through the polarization controller (PC1), are transmitted over 220 km of TW-RS fiber in a Raman amplified system [5], and are demultiplexed and detected. Before the demultiplexer, some light is tapped off. The tapped fraction passes through the tunable optical bandpass filter which alternatively selects the upper or lower sideband of a given channel. Then the Stokes parameters of light in the sidebands are measured using the polarimeter.

Since all channels emerge from the multiplexer with different polarizations, we could not insert a polarizer before PC1 in Fig. 1(a) in our measurement setup. Therefore, in order to make

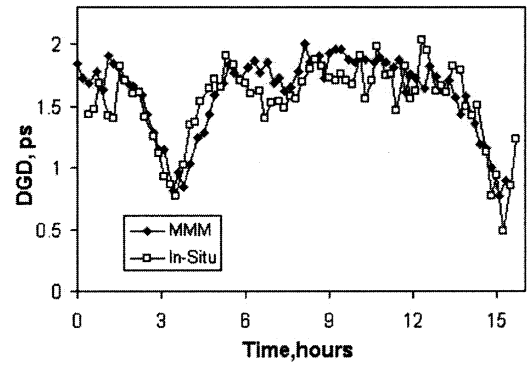


(a)

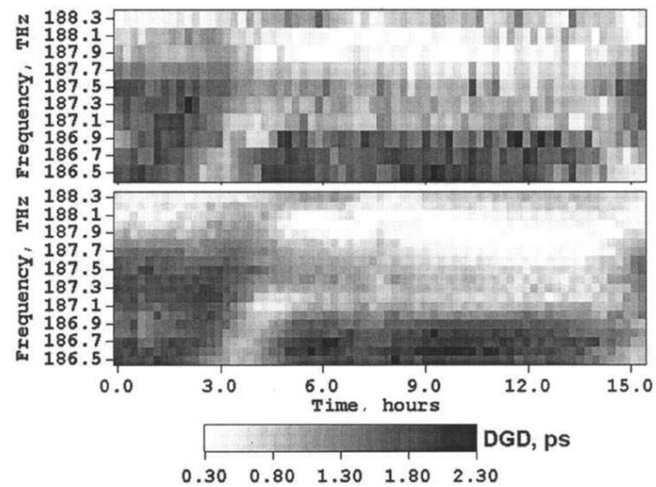


(b)

Fig. 1. (a) Experimental setup consists of the computer-controlled polarization controller (PC1), bandpass filter, and a polarimeter. MMM measurement setup is shown in dashed lines. (b) Optical spectrum of CSRZ signal. Shaded areas correspond to the two bandpass filter positions used for selecting individual sidebands.



(a)



(b)

Fig. 2. Comparison of two measurement methods on a 220-km link. The measurements at each frequency were taken with 20-s offset, which is much less than the time correlation. (a) DGD evolution of a single channel. (b) DGD spectrum evolution (top: *in situ*, bottom: MMM).

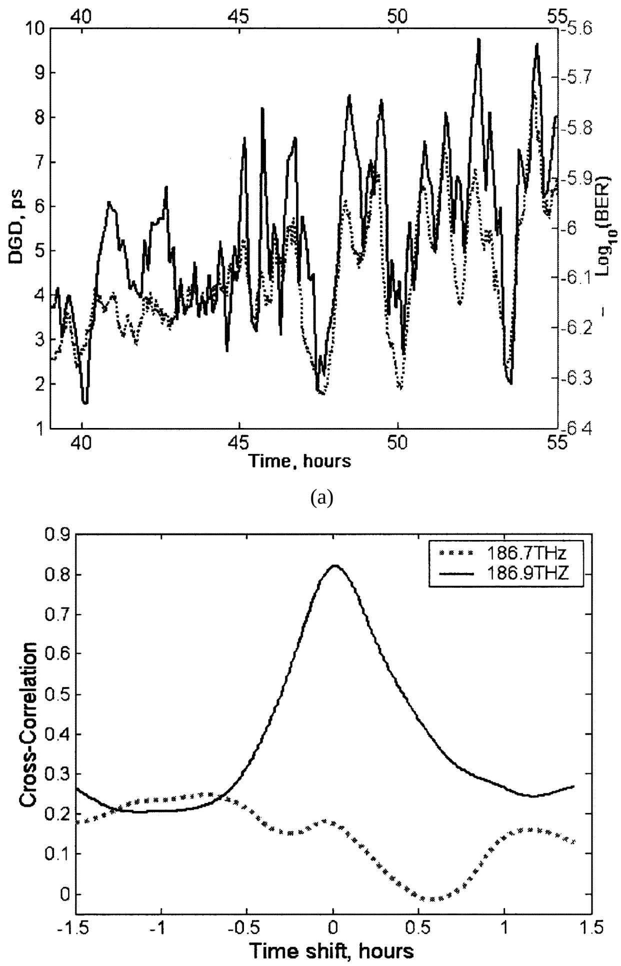
sure that at least two significantly different SOPs are launched into the system, we measured the output SOPs for three launch polarizations for every frequency. Based on three pairs of launched polarizations, we calculated three values of DGD, and then used the mean of the two closest values. By proper choice of the three positions of the PC1, one can ensure that at least two out of three launch polarizations are significantly different for all channels.

In the lower left-hand part of Fig. 1(a), the standard PMD measurement apparatus is shown with dashed lines. This setup, nearly identical to the one described in [6], was used to validate the measurements taken with the *in situ* method. An accurate measurement by the classical MMM method requires the launch state of polarization to be well-controlled. For this purpose, there is a polarizer preceding PC2 which is used to set the launch SOP.

To check the validity of the *in situ* measurement technique in a system with rapidly changing PMD, we time-interleaved measurements by two methods on a 220-km Raman-amplified link. The MMM and *in situ* measurements at each frequency were taken with a 20-s offset, which is much less than the independently calculated time correlation (1 h for 1000-km link, and even longer for 220 km). The *in situ* measurements were performed using the sidebands of each of the eight 200-GHz-spaced 43-Gb/s channels. The probe laser measurements were done on a 100-GHz grid, arranged so that each data channel was bracketed by two probe frequencies, one 50 GHz above the data channel and one 50 GHz below.

Time evolution of DGD measured by these two methods for one of the channels is shown in Fig. 2(a). Shown in Fig. 2(b) is the time evolution of PMD over the entire spectrum. Each frequency scan took 12 min. Since the *in situ* method measures the PMD on the center frequency of the channel, the data obtained with the probe is more detailed. However, one can see that the spectrum evolution trends are essentially the same. Now, with a validated method for *in situ* PMD measurement, we can simultaneously measure PMD and the BER in an operating system. The following measurement is performed on a 1000-km Raman amplified link, similar to [5], with average $\mathrm{DGD}\langle\tau\rangle = 4$ ps. BERs for one of the channels are taken from the number of bytes corrected by the forward-error-correcting (FEC) code and were recorded at approximately 10-s intervals. Meanwhile, *in situ* DGD measurements took about 20 s per wavelength. Since we launched the channels at three different polarizations, with no specific knowledge about the mutual orientation of the SOPs or the PSPs, we used the average of log (BER) as a rough measure of the link quality. Even though the PMD-induced penalty was small, we expected to see at least some correlation between instantaneous BERs and DGD [see Fig. 3(a)]. To make computations easier, we shifted and normalized all functions as follows:

$$\mathrm{BER}_i \to \frac{\mathrm{BER}_i - \langle \mathrm{BER}_i \rangle}{\mathrm{stdev}(\mathrm{BER}_i)}; \quad \tau_j \to \frac{\tau_j - \langle \tau_j \rangle}{\mathrm{stdev}(\tau_j)} \tag{1}$$



(a)

(b)

Fig. 3. (a) Portion of time record of DGD and BER for 186.9-THz channel. (b) Cross correlation between BER for 186.9-THz channel and DGDs for 186.9 THz and adjacent 186.7 channel.

where i, j are channel numbers and τ is DGD. Then the cross-correlation functions $C_{i,j}(t_0)$ were computed between the normalized BER for channel i and a DGD for channel j

$$C_{i,j}(t_0) \rightarrow \frac{1}{T_2 - T_1} \int_{T_1}^{T^2} \mathrm{BER}_i(t) \cdot \tau_j(t - t_0)dt. \quad (2)$$

Plotted in Fig. 3(b) are the cross-correlation functions between the BER for one of the channels (186.9 THz) with the DGD for that same channel (black line) and with the DGD of an adjacent -186.7-THz channel for illustration (dotted line). With a maximum cross correlation of 0.82 BER and DGD for the same channels are well correlated, but there is little correlation between BER and DGD for adjacent channels. This is an expected result, since the 200-GHz-channel separation is much larger than the ~30-GHz correlation bandwidth expected [1] for a link with average DGD $\langle\tau\rangle = 4$ ps. Fig. 3(a) illustrates that the good correlation between the DGD data and the BER data can be seen with the naked eye.

Such a correlation allows us to look at the data from a different point of view; we can see how much variation in the Q value exists in the system for each DGD value. These variations can be caused by power fluctuations, polarization-dependent loss, or by higher orders of PMD. We plot the instantaneous Q values determined from the measured BERs versus the instantaneous DGD for the same channel in Fig. 4. To guide the eye, we also plot the moving average of Q, and the width of the distribution of Qs for each DGD window.

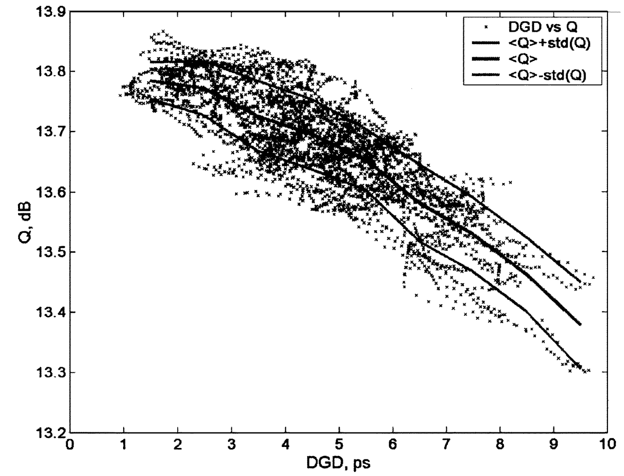


Fig. 4. Scatter plot of the Q factor versus instantaneous DGD. A distribution of Q factors for a given DGD value is evident, as is a correlation.

III. Discussion and Conclusion

We have presented a novel method for nondisruptive PMD measurement in high-speed wavelength-division-multiplexing systems which only requires a single PC (but no polarizer), bandpass filter, and a polarimeter. This method was tested by a simultaneous comparison to a standard MMM method, and applied to studying the correlation between BER and DGD in an installed Raman-amplified 40-Gb/s 1000-km system. The proposed method can be extended to other modulation formats.

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